

# COMPUTING THE PROPORTION OF TRANSITIVE PERMUTATION GROUPS WITH ABELIAN CENTRALIZER

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ABSTRACT. We study the centralizer  $C_{S_d}(G)$  of a transitive permutation group. The centralizer-normalizer formula reduces the question to  $N_G(G_1)/G_1$ , where  $G_1$  is the stabilizer of 1 in  $G$ . Although most of the centralizers are abelian, our computations reveal two different types of nonabelian centralizers: regular nonabelian groups and nonregular stabilizer-normalizer exceptions. The former cases are easier to find, while the latter cases are more mysterious. We compute such nonabelian centralizers in small degrees, prove the first nonregular occurrence at  $d = 12$  together with all of its nonabelian centralizer appearances, and analyze the jump at  $d = 16$ .

## 1. INTRODUCTION

Let  $S_d$  denote the symmetric group on  $d$  points and let  $G \leq S_d$  be a transitive subgroup. The centralizer of  $G$  in  $S_d$  is

$$C_{S_d}(G) = \{\sigma \in S_d : \sigma g = g\sigma \text{ for all } g \in G\}.$$

It records the permutations of the  $d$  points that commute with every element of  $G$ , and therefore measures the residual symmetry of the permutation action. Regular actions provide a basic source of examples: the centralizer of a regular permutation group is isomorphic to the group itself. At the opposite extreme, highly transitive actions such as the natural action of  $S_d$  on  $d$  points have trivial centralizer.

This paper investigates the following quantitative question: how often is  $C_{S_d}(G)$  abelian as  $G$  ranges over transitive subgroups of  $S_d$ ? Concretely, define

$$N_d = \#(\{G \leq S_d : G \text{ is transitive}\} / S_d\text{-conjugacy}),$$

and

$$A_d = \#(\{G \leq S_d : G \text{ is transitive and } C_{S_d}(G) \text{ is abelian}\} / S_d\text{-conjugacy}).$$

Equivalently, two transitive subgroups  $G_1, G_2 \leq S_d$  are identified if there exists  $\tau \in S_d$  such that

$$G_2 = \tau G_1 \tau^{-1}.$$

This is stronger than merely identifying abstractly isomorphic groups, and with this convention  $N_d$  agrees with the number of degree- $d$  transitive groups in the GAP `TransitiveGroups` library.

Our principal object of study is the ratio

$$\frac{A_d}{N_d} \in (0, 1],$$

together with a structural description of the exceptional set  $N_d - A_d$ , namely the transitive groups whose centralizer in  $S_d$  is nonabelian.

**Motivation from Hurwitz spaces.** One motivation for this question comes from the classical construction of Hurwitz spaces using Nielsen classes and braid group actions. Let  $G \leq S_d$  be a fixed transitive permutation group, and let  $\mathbf{C} = (C_1, \dots, C_n)$  be prescribed conjugacy classes in  $G$ . A Nielsen class consists of tuples

$$\mathbf{g} = (g_1, \dots, g_n) \in G^n$$

such that

$$g_i \in C_i, \quad \langle g_1, \dots, g_n \rangle = G, \quad g_1 \cdots g_n = 1.$$

Variants of this definition allow the conjugacy classes  $C_i$  to be permuted. The product-one condition records the relation among local monodromies of a branched cover of  $\mathbb{P}_{\mathbb{C}}^1$ , while the generation condition ensures that the corresponding cover is connected. The braid group  $B_n$ , equivalently the fundamental group of the unordered configuration space  $\text{UConf}_n(\mathbb{C})$ , acts on these tuples by the classical Artin action. This action is one of the standard combinatorial inputs in the construction of Hurwitz spaces [7, 8, 9].

Geometrically, a point of a Hurwitz space represents a branched cover

$$f : Y \rightarrow \mathbb{P}_{\mathbb{C}}^1$$

with prescribed local monodromy. After choosing a base point away from the branch locus and labeling a generic fiber, the cover determines a transitive monodromy group  $G \leq S_d$ . The centralizer

$$C_{S_d}(G) = \{\sigma \in S_d : \sigma g = g\sigma \text{ for all } g \in G\}$$

records the permutations of the fiber that commute with the monodromy action. Equivalently, in the connected-cover setting, it describes the deck transformations, or automorphisms of the cover over the base, as seen through the monodromy representation.

Thus the condition  $C_{S_d}(G) = 1$  corresponds to the case where the cover has no nontrivial automorphisms over the base. In moduli problems, however, covers with nontrivial automorphisms cannot simply be ignored; indeed, they are one reason Hurwitz stacks rather than only coarse Hurwitz spaces naturally appear. This makes it useful to understand not only when  $C_{S_d}(G)$  is trivial, but also when it is abelian or nonabelian.

This leads to the finite computational problem considered here. For each degree  $d$ , we compute the ratio

$$\frac{A_d}{N_d},$$

identify the exceptional cases with nonabelian centralizer, separate the structural mechanisms that produce them, and use the resulting data to formulate further questions about the dominance of the abelian-centralizer case.

**Computational and group-theoretic motivation.** The same question also has a direct computational and group-theoretic motivation. It can be viewed as an analogue of classical commuting-probability questions, in which one asks how frequently two random elements of a finite group commute. Here, instead of asking whether two individual elements commute, we ask how frequently the entire commutant of a transitive subgroup is abelian.

The question is especially tractable because the centralizer of a transitive permutation group has a transparent description in terms of point stabilizers and normalizers. If  $G \leq S_d$  is transitive and  $G_1$  is the stabilizer of 1, then Proposition 3.1 gives

$$C_{S_d}(G) \cong N_G(G_1)/G_1.$$

Thus the problem is not merely to count exceptions by computer, but to understand which internal stabilizer-normalizer structures give rise to non-abelian centralizers.

**Main contributions.** The main contributions of this paper are as follows.

- (1) We tabulate  $N_d$ ,  $A_d$ , and  $A_d/N_d$  for every  $4 \leq d \leq 30$ , with the results split across Tables 1 and 2.
- (2) We isolate two structural mechanisms that produce nonabelian centralizers: the regular mechanism and the stabilizer-normalizer mechanism. In the regular mechanism,  $G$  itself is a regular nonabelian permutation group, so  $C_{S_d}(G) \cong G$ . In the stabilizer-normalizer mechanism,  $G$  is nonregular but  $N_G(G_1)/G_1$  is nonabelian.
- (3) We give a complete case-by-case analysis of  $d = 12$ , the smallest degree at which the stabilizer-normalizer mechanism produces non-regular examples.
- (4) We analyze the degree  $d = 16$ , where the regular mechanism gives nine nonabelian-centralizer exceptions and the full SageMath enumeration identifies sixteen additional nonregular stabilizer-normalizer exceptions. Thus  $N_{16} - A_{16} = 25$ .
- (5) We prove a prime-degree result showing that if  $p$  is prime and  $G \leq S_p$  is transitive, then  $C_{S_p}(G)$  is abelian; more precisely, it is either trivial or cyclic of order  $p$ .

**Organization.** Section 2 fixes the terminology and notation used throughout the paper, including transitive, regular, primitive, stabilizer, normalizer, and centralizer. Section 3 proves the structural tools behind the computation, especially the centralizer-normalizer formula and its consequences. Section 4 describes the computational method and the optimized SageMath pipeline. Section 5 reports the data for  $4 \leq d \leq 30$  and discusses the general patterns suggested by the tables. Section 6 gives the detailed case study of

$d = 12$ . Section 7 analyzes the jump at  $d = 16$ . Section 8 records several future directions and open questions suggested by the data.

## 2. BACKGROUND AND NOTATION

All groups in this paper are finite. We write

$$\Omega = \{1, 2, \dots, d\}$$

for the  $d$ -element set on which  $S_d$  acts. A permutation group of degree  $d$  is a subgroup  $G \leq S_d$ . We view elements of  $S_d$  as bijections  $\Omega \rightarrow \Omega$ , and group multiplication is read left-to-right:  $(\sigma g)(x)$  means apply  $\sigma$  first and then apply  $g$ .

**2.1. Permutation actions, orbits, and stabilizers.** A group action of a group  $G$  on a set  $\Omega$  is a rule that assigns to each  $g \in G$  a permutation of  $\Omega$ , compatibly with the group operation. In this paper, the action is always given by an inclusion  $G \leq S_d$ , so each element  $g \in G$  acts directly as a permutation of  $\Omega$ .

For  $i \in \Omega$ , the orbit of  $i$  under  $G$  is

$$G \cdot i = \{g(i) : g \in G\}.$$

The group  $G \leq S_d$  is transitive if it has a single orbit on  $\Omega$ ; equivalently, for every pair  $i, j \in \Omega$ , there exists  $g \in G$  such that  $g(i) = j$ .

The stabilizer of  $i \in \Omega$  in  $G$  is

$$G_i = \{g \in G : g(i) = i\}.$$

In particular, throughout the paper we write

$$G_1 = \{g \in G : g(1) = 1\}$$

for the stabilizer of the point 1.

**2.2. Normalizers and centralizers.** Let  $H \leq G$  be a subgroup. The normalizer of  $H$  in  $G$  is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

It is the largest subgroup of  $G$  in which  $H$  is normal.

If  $G \leq S_d$ , the centralizer of  $G$  in  $S_d$  is

$$C_{S_d}(G) = \{\sigma \in S_d : \sigma g = g\sigma \text{ for all } g \in G\}.$$

Thus  $C_{S_d}(G)$  consists of the permutations of  $\Omega$  that commute with every element of  $G$ .

**2.3. A basic semiregularity lemma.** The following elementary observation is used repeatedly.

**Lemma 2.1.** *Let  $G \leq S_d$  be transitive. Then  $C_{S_d}(G)$  acts semiregularly on  $\Omega$ . Equivalently, if  $\sigma \in C_{S_d}(G)$  fixes one point of  $\Omega$ , then  $\sigma$  is the identity permutation.*

*Proof.* Suppose  $\sigma \in C_{S_d}(G)$  fixes a point  $i \in \Omega$ . We show that  $\sigma$  fixes every point of  $\Omega$ . Let  $j \in \Omega$ . Since  $G$  is transitive, there exists  $g \in G$  such that  $g(i) = j$ . Because  $\sigma$  commutes with  $g$ , we have

$$\sigma(j) = \sigma(g(i)) = g(\sigma(i)) = g(i) = j.$$

Thus  $\sigma$  fixes every  $j \in \Omega$ , so  $\sigma$  is the identity.  $\square$

**2.4. Regular, semiregular, and primitive actions.** An action of  $G \leq S_d$  is semiregular if every nonidentity element of  $G$  fixes no point of  $\Omega$ . Equivalently,  $G_i = \{1\}$  for every  $i \in \Omega$ .

A transitive action is regular if it is both transitive and semiregular. Equivalently, for a transitive group  $G \leq S_d$ , the following conditions are equivalent:

$$G \text{ is regular,} \quad G_1 = \{1\}, \quad |G| = d.$$

A regular subgroup of  $S_d$  is the image of the left or right regular representation of an abstract group of order  $d$ .

A transitive action  $G \leq S_d$  is primitive if  $\Omega$  admits no nontrivial  $G$ -invariant partition. Equivalently, the only  $G$ -invariant partitions of  $\Omega$  are the partition into singletons and the one-block partition  $\{\Omega\}$ . If a transitive action is not primitive, it is called imprimitive.

**Proposition 2.1.** *Let  $G \leq S_d$  be primitive. Then  $C_{S_d}(G)$  is abelian.*

*Proof.* Let

$$C = C_{S_d}(G).$$

The orbits of  $C$  on  $\Omega$  form a  $G$ -invariant partition of  $\Omega$ . Indeed, if  $x, y \in \Omega$  lie in the same  $C$ -orbit, then there exists  $c \in C$  such that  $c(x) = y$ . For any  $g \in G$ , since  $c$  commutes with  $g$ , we have

$$c(g(x)) = g(c(x)) = g(y),$$

so  $g(x)$  and  $g(y)$  again lie in the same  $C$ -orbit.

Since  $G$  is primitive, this partition is either the partition into singletons or the one-block partition  $\{\Omega\}$ . In the first case,  $C$  is trivial, hence abelian. In the second case,  $C$  is transitive. By Lemma 2.1,  $C = C_{S_d}(G)$  acts semiregularly on  $\Omega$ . Hence  $C$  acts regularly.

It remains to show that this regular group  $C$  is abelian. Since  $C = C_{S_d}(G)$ , every element of  $G$  commutes with every element of  $C$ , so

$$G \leq C_{S_d}(C).$$

Since  $C$  is regular, the centralizer  $C_{S_d}(C)$  is also regular on  $\Omega$ . Hence

$$|C_{S_d}(C)| = d.$$

Since  $G$  is transitive,  $|G| \geq d$ . Therefore

$$d \leq |G| \leq |C_{S_d}(C)| = d,$$

so

$$G = C_{S_d}(C).$$

Now suppose  $K \leq C$  is a nontrivial proper subgroup. Since every element of  $C$  commutes with every element of  $G$ , the orbits of  $K$  on  $\Omega$  form a  $G$ -invariant partition of  $\Omega$ . Because  $C$  is regular, the  $K$ -orbits are neither singletons nor all of  $\Omega$ . Thus they form a nontrivial block system for  $G$ , contradicting the primitivity of  $G$ . Hence  $C$  has no nontrivial proper subgroups. By Cauchy's theorem, this forces  $|C|$  to be prime. Therefore  $C$  is cyclic, and in particular  $C$  is abelian.  $\square$

**Lemma 2.2.** *Let  $p$  be prime and let  $G \leq S_p$  be transitive. Then  $G$  is primitive.*

*Proof.* Suppose  $G$  admits a nontrivial block system on  $\Omega = \{1, \dots, p\}$ . Since  $G$  is transitive, all blocks in a block system have the same size. Thus the block size must divide  $p$ . Because  $p$  is prime, the only possible block sizes are 1 and  $p$ . These give only the partition into singletons and the one-block partition  $\{\Omega\}$ . Hence there is no nontrivial block system, so  $G$  is primitive.  $\square$

**Theorem 2.1.** *Let  $p$  be prime and let  $G \leq S_p$  be transitive. Then  $C_{S_p}(G)$  is abelian. More precisely,  $C_{S_p}(G)$  is either trivial or cyclic of order  $p$ .*

*Proof.* By Lemma 2.2,  $G$  is primitive. Therefore Proposition 2.1 implies that  $C_{S_p}(G)$  is abelian.

Moreover, since  $G$  is transitive, Lemma 2.1 shows that  $C_{S_p}(G)$  acts semiregularly on the  $p$ -point set. Hence

$$|C_{S_p}(G)| \mid p.$$

Since  $p$  is prime, either  $|C_{S_p}(G)| = 1$  or  $|C_{S_p}(G)| = p$ . In the first case the centralizer is trivial. In the second case it has prime order, so it is cyclic of order  $p$ .  $\square$

Up to conjugacy in  $S_d$ , the transitive subgroups of small degree have been classified and tabulated by Hulpke [1]. We use his numbering throughout, as implemented by `TransitiveGroup(d,k)` in GAP.

**2.5. Direct, semidirect, and wreath products.** We recall three standard group constructions that appear in the tables below.

The direct product  $G \times H$  is the group whose elements are ordered pairs  $(g, h)$ , with componentwise multiplication:

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2).$$

Let  $N$  and  $H$  be groups, and let  $\varphi : H \rightarrow \text{Aut}(N)$  be a homomorphism. The semidirect product  $N \rtimes_{\varphi} H$  is the group with underlying set  $N \times H$  and multiplication

$$(n_1, h_1)(n_2, h_2) = (n_1\varphi(h_1)(n_2), h_1h_2).$$

When the action  $\varphi$  is understood, we write  $N \rtimes H$ . Examples appearing below include  $C_3 \rtimes C_4$  and  $(C_3 \times C_3) \rtimes C_4$ .

Let  $G$  act on a set  $\Delta$ , and let  $H$  act on a set  $\Theta$  of size  $n$ . The wreath product  $G \wr H$  is the semidirect product

$$G^n \rtimes H,$$

where  $H$  permutes the  $n$  copies of  $G$ . In its imprimitive product action on  $\Delta \times \Theta$ , the wreath product embeds into  $S_{|\Delta|n}$ .

**2.6. Centralizer of a transitive abelian group.** The following classical fact is used as a basic reference point.

**Proposition 2.2.** *If  $G \leq S_d$  is transitive and abelian, then  $C_{S_d}(G) = G$ . In particular,  $G$  is regular.*

*Proof.* Since  $G$  is abelian, every element of  $G$  commutes with every other element of  $G$ , so

$$G \leq C_{S_d}(G).$$

By the semiregularity lemma above,  $C_{S_d}(G)$  acts semiregularly on  $\Omega$ . Hence

$$|C_{S_d}(G)| \leq d.$$

On the other hand, since  $G$  is transitive, the orbit-stabilizer theorem gives

$$|G| = d|G_1|,$$

so  $d \mid |G|$ , and therefore  $|G| \geq d$ . Combining

$$G \leq C_{S_d}(G), \quad |C_{S_d}(G)| \leq d, \quad |G| \geq d,$$

we get

$$|G| = |C_{S_d}(G)| = d.$$

Therefore  $G = C_{S_d}(G)$ , and  $G$  is regular. □

### 3. STRUCTURAL TOOLS

Two propositions reduce the computation of  $C_{S_d}(G)$  to questions purely internal to  $G$ .

#### 3.1. The centralizer–normalizer formula.

**Proposition 3.1.** *Let  $G \leq S_d$  be transitive and let  $H = G_1$  be the stabilizer of 1. Then*

$$C_{S_d}(G) \cong N_G(H)/H.$$

*Sketch.* Identify  $\Omega$  with the coset space  $G/H$  via  $i \leftrightarrow g_i H$ , where  $g_i \in G$  is any element sending 1 to  $i$  (such  $g_i$  exist by transitivity, and the coset  $g_i H$  is independent of the choice). Under this identification,  $G$  acts on  $G/H$  by left multiplication. A permutation  $\sigma \in S_d$  commutes with this action iff  $\sigma$  comes from right multiplication by an element of  $N_G(H)$  modulo  $H$ . The map

$$nH \mapsto (gH \mapsto gn^{-1}H)$$

gives the asserted isomorphism. □

### 3.2. Characterization of the regular case.

**Proposition 3.2.** *Let  $G \leq S_d$  be transitive. Then  $G$  is regular if and only if*

$$C_{S_d}(G) \cong G.$$

*Proof.* First suppose that  $G$  is regular. Then  $G_1 = \{1\}$ . By Proposition 3.1,

$$C_{S_d}(G) \cong N_G(G_1)/G_1.$$

Since  $G_1 = \{1\}$ , we have  $N_G(G_1) = G$ , and therefore

$$C_{S_d}(G) \cong G/\{1\} \cong G.$$

Conversely, suppose that  $G \leq S_d$  is transitive and

$$C_{S_d}(G) \cong G.$$

By Lemma 2.1,  $C_{S_d}(G)$  acts semiregularly on  $\Omega$ , so

$$|C_{S_d}(G)| \leq d.$$

Since  $G$  is transitive, the orbit-stabilizer theorem gives

$$|G| = d|G_1|,$$

so  $|G| \geq d$ . The assumption  $C_{S_d}(G) \cong G$  gives

$$|C_{S_d}(G)| = |G|.$$

Therefore

$$d \leq |G| = |C_{S_d}(G)| \leq d.$$

Hence  $|G| = d$ . Since  $G$  is transitive, the equality

$$|G| = d|G_1|$$

implies  $|G_1| = 1$ . Thus  $G$  is regular. □

### 3.3. A divisibility corollary.

**Corollary 3.1.** *For every transitive  $G \leq S_d$ ,*

$$|C_{S_d}(G)| \mid d.$$

*In particular, every regular nonabelian subgroup of  $S_d$  contributes one nonabelian-centralizer exception, and for every  $d$  at which a nonabelian group of order  $d$  exists,*

$$N_d - A_d \geq \#\{\text{nonabelian groups of order } d\}.$$

*Proof.* By Proposition 3.1,

$$|C_{S_d}(G)| = |N_G(G_1)/G_1| = \frac{|N_G(G_1)|}{|G_1|}.$$

Since  $N_G(G_1) \leq G$ , this divides

$$\frac{|G|}{|G_1|} = d.$$



Hence

$$|C_{S_d}(G)| \mid d.$$

For the second statement, suppose  $G$  is a regular nonabelian subgroup of  $S_d$ . By Proposition 3.2,

$$C_{S_d}(G) \cong G.$$

Thus  $C_{S_d}(G)$  is nonabelian. Two regular subgroups isomorphic to the same abstract group  $G_0$  are  $S_d$ -conjugate, since any two regular permutation representations of  $G_0$  on the same  $d$ -element set are conjugate. Therefore distinct abstract nonabelian groups of order  $d$  contribute distinct  $S_d$ -conjugacy classes of transitive subgroups with nonabelian centralizer, giving the stated lower bound.  $\square$

*Remark 3.1.* Corollary 3.1 is useful not only as a divisibility statement, but also as a guide for reducing the computational search. If  $G \leq S_d$  is transitive and

$$H \cong C_{S_d}(G),$$

then Corollary 3.1 implies that  $|H| \mid d$ . Write

$$d = m|H|.$$

Equivalently,

$$m = \frac{d}{|C_{S_d}(G)|}.$$

Thus, to search for examples with a prescribed nonabelian centralizer  $H$ , one may first fix a divisor  $m$  of  $d$ , list the possible nonabelian groups  $H$  of order  $d/m$ , and then search for transitive subgroups  $G \leq S_d$  whose centralizer is isomorphic to  $H$ .

This can reduce the search space. Since  $C_{S_d}(G)$  acts semiregularly on  $\Omega$ , its orbits have size  $|H|$ , and there are  $m$  such orbits. The commuting relation between  $G$  and  $C_{S_d}(G)$  forces the action of  $G$  to respect this orbit decomposition up to permutation of the  $m$  orbits. Thus, in favorable cases, the search may be restricted from all of  $S_d$  to a wreath-product-type subgroup of the form

$$H \wr S_m \cong H^m \rtimes S_m \leq S_d.$$

The usefulness of this reduction depends on the implementation, but it explains why Corollary 3.1 can serve as a pruning tool in the search for nonabelian centralizers.

The regular mechanism is the extreme case  $m = 1$ . In that case

$$|C_{S_d}(G)| = d,$$

and Proposition 3.2 gives

$$C_{S_d}(G) \cong G.$$

Therefore regular nonabelian exceptions are obtained simply by listing the nonabelian groups of order  $d$ . For example, the regular exceptions at degree  $d = 12$  come directly from the nonabelian groups of order 12.

## 4. COMPUTATIONAL METHOD

For each degree  $d$ , we use SageMath, which wraps GAP's `TransitiveGroups` library, to enumerate the transitive subgroups of  $S_d$  up to conjugacy. For each transitive group  $G \leq S_d$ , we let  $G_1$  denote the stabilizer of the point 1. By Proposition 3.1,

$$C_{S_d}(G) \cong N_G(G_1)/G_1.$$

Thus the problem of deciding whether  $C_{S_d}(G)$  is abelian can be replaced by the internal group-theoretic computation of the quotient  $N_G(G_1)/G_1$ .

**4.1. Naive algorithm and its bottleneck.** The most direct algorithm is to compute the normalizer quotient  $N_G(G_1)/G_1$  for every transitive group  $G$  in degree  $d$ .

```
for each d in 4..D_MAX:
  T <- TransitiveGroups(d)
  N_d <- |T|
  A_d <- 0
  exceptions <- []

  for each G in T:
    H <- Stabilizer(G, 1)
    Q <- Normalizer(G, H) / H

    if IsAbelian(Q):
      A_d <- A_d + 1
    else:
      exceptions.append((G, |H|, |N_G(H)|, Q))

  record(d, N_d, A_d, exceptions)
```

This algorithm follows Proposition 3.1 directly. Its advantage is conceptual simplicity: every group  $G$  is treated in the same way, and every centralizer is detected through the quotient  $N_G(G_1)/G_1$ .

The bottleneck is the repeated computation of  $\text{Normalizer}(G, G_1)$ . If  $N_d$  denotes the number of transitive subgroups of  $S_d$  in the GAP library, then the naive algorithm performs  $N_d$  normalizer computations in degree  $d$ . The cost of each normalizer computation depends strongly on the size and structure of  $G_1$ . In practice, the computation becomes slowest for imprimitive nonregular groups with large point stabilizers. Thus the total running time is dominated not by the enumeration of the transitive groups, but by the repeated normalizer calls.

The space cost is comparatively modest for small degrees: the algorithm stores the current group, its point stabilizer, the quotient  $N_G(G_1)/G_1$ , and the list of nonabelian exceptions. However, as  $d$  increases, the number  $N_d$  of transitive groups can grow rapidly, and the exception list together with

the group data returned by GAP can become large. The main practical obstruction is still time rather than space in the range considered here.

**4.2. Optimized algorithm.** The optimized algorithm avoids the normalizer computation whenever a simpler structural test already decides whether the centralizer is abelian. There are two such cases.

First, if  $G$  is primitive, then Proposition 2.1 implies that  $C_{S_d}(G)$  is abelian. Thus primitive groups can be counted immediately.

Second, if  $G$  is regular, then Proposition 3.2 gives

$$C_{S_d}(G) \cong G.$$

Therefore a regular group contributes an abelian centralizer exactly when  $G$  itself is abelian. This avoids a normalizer computation in the regular case as well.

The optimized algorithm is therefore:

```
for each d in 4..D_MAX:
  T <- TransitiveGroups(d)
  N_d <- |T|
  A_d <- 0
  exceptions <- []

  for each G in T:

    if IsPrimitive(G):
      A_d <- A_d + 1
      continue

    H <- Stabilizer(G, 1)

    if |H| = 1:
      if IsAbelian(G):
        A_d <- A_d + 1
      else:
        exceptions.append((G, 1, |G|, G))
        continue

    Q <- Normalizer(G, H) / H

    if IsAbelian(Q):
      A_d <- A_d + 1
    else:
      exceptions.append((G, |H|, |N_G(H)|, Q))

record(d, N_d, A_d, exceptions)
```

Compared with the naive algorithm, the optimized algorithm calls  $\text{Normalizer}(G, G_1)$  only for groups that are both nonprimitive and nonregular. Thus the number of expensive normalizer calls is reduced from

$$N_d$$

to

$$\#\{G \leq S_d : G \text{ is transitive, nonprimitive, and nonregular}\},$$

with all counts taken up to  $S_d$ -conjugacy.

The time saved depends on the degree. The improvement is largest in degrees where many transitive groups are primitive or regular. It is smaller in degrees where most groups are imprimitive and nonregular, since those cases still require the normalizer quotient  $N_G(G_1)/G_1$ .

The space complexity is essentially unchanged from the naive algorithm: the optimized version stores the same type of data, but it performs fewer expensive quotient computations. Its main advantage is therefore a reduction in running time rather than a reduction in memory usage.

**4.3. Search reduction using centralizer size.** Corollary 3.1 gives an additional way to organize the search. Since

$$|C_{S_d}(G)| \mid d$$

for every transitive  $G \leq S_d$ , any possible nonabelian centralizer must have order dividing  $d$ . If

$$H \cong C_{S_d}(G),$$

then we may write

$$d = m|H|.$$

Equivalently,

$$m = \frac{d}{|C_{S_d}(G)|}.$$

Thus one can first choose a divisor  $m$  of  $d$ , then list possible nonabelian groups  $H$  of order  $d/m$ , and finally search for transitive groups  $G \leq S_d$  whose centralizer is isomorphic to  $H$ .

This observation can prune the search. Since  $C_{S_d}(G)$  acts semiregularly on  $\Omega$ , its orbits have size  $|H|$ , and there are  $m$  such orbits. The commuting relation between  $G$  and  $C_{S_d}(G)$  forces the action of  $G$  to respect this orbit decomposition up to permutation of the  $m$  orbits. Thus, in favorable cases, the search may be restricted from all of  $S_d$  to a wreath-product-type subgroup of the form

$$H \wr S_m \cong H^m \rtimes S_m \leq S_d.$$

The usefulness of this reduction depends on the implementation, but it explains why Corollary 3.1 can serve as a pruning tool in the search for nonabelian centralizers.

The regular mechanism is the extreme case  $m = 1$ . In this case

$$|C_{S_d}(G)| = d,$$

and Proposition 3.2 gives

$$C_{S_d}(G) \cong G.$$

Therefore regular nonabelian exceptions are obtained simply by listing the nonabelian groups of order  $d$ .

**4.4. Implementation used in the experiments.** The computations reported in this paper use the optimized algorithm above. The primitive and regular cases are filtered before calling `Normalizer( $G, G_1$ )`. The normalizer quotient is computed only for the remaining nonprimitive nonregular groups.

For every group flagged as a nonabelian-centralizer exception, we record the degree  $d$ , the GAP transitive group identifier, the order of  $G$ , the order of the point stabilizer  $G_1$ , the order of  $N_G(G_1)$ , and the quotient  $N_G(G_1)/G_1$ . This information is used in the tables below to distinguish the regular mechanism from the stabilizer-normalizer mechanism.

The optimized algorithm is sufficient for the degrees considered here. However, it does not remove the main difficulty entirely: for larger degrees, especially those with many imprimitive nonregular transitive groups, the normalizer computations can still dominate the running time. A more refined implementation using the search reduction described above could further prune the candidate space, but we leave a systematic implementation of that idea for future work.

## 5. COMPUTATIONAL RESULTS FOR $4 \leq d \leq 30$

Table 1 reports  $N_d$ ,  $A_d$ , the difference, the ratio  $A_d/N_d$ , and the nonabelian exceptions for  $4 \leq d \leq 15$ .

Table 2 extends the computation to  $16 \leq d \leq 30$ . Every entry in Table 2 has been independently verified by a SageMath re-run on the GAP `TransitiveGroups` library using the optimised pipeline of Listing 2; the extra columns “reg” and “SN” record how many exceptions arise from the regular mechanism (Proposition 3.2) versus the stabilizer-normalizer mechanism (Proposition 3.1 with  $|G_1| > 1$ ).

Two empirical patterns deserve immediate comment. First, the prime rows ( $d \in \{17, 19, 23, 29\}$ ) all give  $A_d/N_d = 1$ , consistent with Theorem 2.1. Second, the rows at  $d = p^2$  in our range ( $p = 2$ :  $d = 4$ ,  $p = 3$ :  $d = 9$ ,  $p = 5$ :  $d = 25$ ) also all give ratio 1, suggesting an unconditional  $A_{p^2} = N_{p^2}$  statement for odd primes  $p$ ; we revisit this as Question Q5 in Section 8. The single striking outlier in the extended range is  $d = 24$  with 163 exceptions, of which only 12 come from the regular mechanism; the remaining 151 are stabilizer-normalizer cases, by a wide margin the densest pocket of SN exceptions we have observed.

**5.1. General pattern.** Across the entire range  $4 \leq d \leq 30$ , the ratio  $A_d/N_d$  never drops below  $1415/1428 \approx 0.991$  (attained at  $d = 30$ ). For thirteen of the twenty-seven degrees the ratio equals exactly 1. The exceptions, when they exist, are sparse and exhibit visible structure rather than appearing

TABLE 1. Counts of transitive groups of degree  $4 \leq d \leq 15$  and their abelian-centralizer ratios.

| $d$ | $N_d$ | $A_d$ | $N_d - A_d$ | $A_d/N_d$ | nonabelian exceptions |
|-----|-------|-------|-------------|-----------|-----------------------|
| 4   | 5     | 5     | 0           | 1         | –                     |
| 5   | 5     | 5     | 0           | 1         | –                     |
| 6   | 16    | 15    | 1           | 15/16     | $S_3$                 |
| 7   | 7     | 7     | 0           | 1         | –                     |
| 8   | 50    | 48    | 2           | 24/25     | $D_4, Q_8$            |
| 9   | 34    | 34    | 0           | 1         | –                     |
| 10  | 45    | 44    | 1           | 44/45     | $D_5$                 |
| 11  | 8     | 8     | 0           | 1         | –                     |
| 12  | 301   | 295   | 6           | 295/301   | see Table 3           |
| 13  | 9     | 9     | 0           | 1         | –                     |
| 14  | 63    | 62    | 1           | 62/63     | $D_7$                 |
| 15  | 104   | 104   | 0           | 1         | –                     |

TABLE 2. Extension of Table 1 to  $16 \leq d \leq 30$ . Every row was independently re-verified by a SageMath run using the optimised pipeline of Listing 2; the “reg” and “SN” columns split the exception count into regular-mechanism and stabilizer-normalizer-mechanism cases. All wall-times are single-core on a 2024-era laptop.

| $d$ | $N_d$ | $A_d$ | $N_d - A_d$ | reg | SN  | $A_d/N_d$   | runtime |
|-----|-------|-------|-------------|-----|-----|-------------|---------|
| 16  | 1954  | 1929  | 25          | 9   | 16  | 1929/1954   | 11.1 s  |
| 17  | 10    | 10    | 0           | 0   | 0   | 1           | 0.0 s   |
| 18  | 983   | 967   | 16          | 3   | 13  | 967/983     | 6.3 s   |
| 19  | 8     | 8     | 0           | 0   | 0   | 1           | 0.0 s   |
| 20  | 1117  | 1111  | 6           | 3   | 3   | 1111/1117   | 8.8 s   |
| 21  | 164   | 163   | 1           | 1   | 0   | 163/164     | 1.0 s   |
| 22  | 59    | 58    | 1           | 1   | 0   | 58/59       | 0.4 s   |
| 23  | 7     | 7     | 0           | 0   | 0   | 1           | 0.0 s   |
| 24  | 25000 | 24837 | 163         | 12  | 151 | 24837/25000 | 225.2 s |
| 25  | 211   | 211   | 0           | 0   | 0   | 1           | 1.2 s   |
| 26  | 96    | 95    | 1           | 1   | 0   | 95/96       | 0.8 s   |
| 27  | 2392  | 2390  | 2           | 2   | 0   | 1195/1196   | 17.2 s  |
| 28  | 1854  | 1849  | 5           | 2   | 3   | 1849/1854   | 17.4 s  |
| 29  | 8     | 8     | 0           | 0   | 0   | 1           | 0.0 s   |
| 30  | 5712  | 5660  | 52          | 3   | 49  | 1415/1428   | 56.3 s  |

at random; the densest pocket is at  $d = 24$  (163 exceptions out of 25,000 transitive groups, still ratio  $\approx 0.993$ ).

**5.2. Prime degrees.** For every prime degree in our range —  $d \in \{5, 7, 11, 13, 17, 19, 23, 29\}$  — we obtain  $A_d/N_d = 1$ . The reason is the prime-degree argument formalised in Theorem 2.1.

*Observation 5.1.* In the computed prime degrees  $d \in \{5, 7, 11, 13, 17, 19, 23, 29\}$ , every transitive subgroup of  $S_d$  has abelian centralizer; moreover this centralizer is cyclic.

*Observation 5.2.* In the computed perfect-square degrees  $d \in \{4, 9, 25\}$ , every transitive subgroup of  $S_d$  has abelian centralizer; in particular  $A_4 = N_4$ ,  $A_9 = N_9$ , and  $A_{25} = N_{25}$ . The contrast with  $d = 16 = 2^4$  (ratio 1929/1954) shows that the relevant property is being the square of an odd prime, which we record as Question Q5 in Section 8.

**5.3. Two mechanisms behind the exceptions.** Every nonabelian exception in Table 1 and Table 2 falls into exactly one of the following structural classes.

- (R) **Regular mechanism.:**  $G$  is regular nonabelian:  $H = G_1 = \{1\}$ , and Proposition 3.2 gives  $C_{S_d}(G) \cong G$ . The exceptions of this type in Table 1 are  $S_3$  ( $d = 6$ ),  $D_4$  and  $Q_8$  ( $d = 8$ ),  $D_5$  ( $d = 10$ ),  $D_7$  ( $d = 14$ ), and (within  $d = 12$ )  $D_6$ ,  $A_4$ ,  $C_3 \rtimes C_4$ . These are precisely the regular permutation representations of the smallest nonabelian groups.
- (SN) **Stabilizer–normalizer mechanism.:**  $G$  is nonregular, so  $H \neq \{1\}$ , but  $N_G(H)/H$  is nonabelian. Here Proposition 3.1 produces a nonabelian centralizer indirectly, via the internal stabilizer–normalizer structure of  $G$  rather than from  $G$  itself. This mechanism first appears at  $d = 12$  and is the entire content of the second half of Table 3.

## 6. CASE STUDY: DEGREE 12

Among the degrees in Table 1,  $d = 12$  is the first degree at which the stabilizer–normalizer mechanism produces nonregular examples. It is also a useful test case because the regular and nonregular mechanisms both occur in a small enough degree to list explicitly.

**Theorem 6.1.** *Among the transitive subgroups of  $S_{12}$ , exactly six have nonabelian centralizer. They are the six groups listed in Table 3. Three of them are regular nonabelian groups of order 12, and the remaining three are nonregular groups for which*

$$N_G(G_1)/G_1 \cong S_3.$$

Consequently,

$$N_{12} - A_{12} = 6, \quad A_{12} = 295, \quad \frac{A_{12}}{N_{12}} = \frac{295}{301}.$$

*Proof.* We use the centralizer–normalizer formula from Proposition 3.1:

$$C_{S_{12}}(G) \cong N_G(G_1)/G_1.$$

First consider the regular cases. If  $G \leq S_{12}$  is regular, then  $|G| = 12$ . By Proposition 3.2,

$$C_{S_{12}}(G) \cong G.$$

Thus a regular subgroup contributes a nonabelian centralizer exactly when the corresponding abstract group of order 12 is nonabelian. There are three nonabelian groups of order 12, namely

$$D_6, \quad A_4, \quad C_3 \rtimes C_4.$$

These give the three regular exceptions in Table 3, with GAP identifiers 3, 4, 5.

It remains to consider the nonregular cases. For every nonregular transitive group  $G \leq S_{12}$ , we computed  $G_1$ ,  $N_G(G_1)$ , and the quotient  $N_G(G_1)/G_1$  using the method described in Section 4. The only nonregular cases for which this quotient is nonabelian are the groups with GAP identifiers 16, 17, 35. In each of these cases the quotient is isomorphic to  $S_3$ :

$$N_G(G_1)/G_1 \cong S_3.$$

More explicitly, for ids 16 and 17 one has

$$|G_1| = 3, \quad |N_G(G_1)| = 18,$$

so

$$|N_G(G_1)/G_1| = 18/3 = 6, \quad N_G(G_1)/G_1 \cong S_3.$$

For id 35 one has

$$|G_1| = 6, \quad |N_G(G_1)| = 36,$$

again giving

$$|N_G(G_1)/G_1| = 36/6 = 6, \quad N_G(G_1)/G_1 \cong S_3.$$

By Proposition 3.1, these three groups therefore have nonabelian centralizer isomorphic to  $S_3$ .

The optimized enumeration described in Section 4 checks all  $N_{12} = 301$  transitive groups of degree 12. All groups not appearing in Table 3 have abelian quotient  $N_G(G_1)/G_1$ , and hence abelian centralizer by Proposition 3.1. Therefore Table 3 gives the complete list of degree-12 nonabelian-centralizer exceptions. The code used for this enumeration is listed among the computational artifacts in Appendix A.

□

The first three rows of Table 3 are the regular-mechanism exceptions. The last three rows are the first nonregular stabilizer-normalizer exceptions in the computed range. Thus  $d = 12$  is the smallest degree at which the SN mechanism appears.



TABLE 3. The six transitive subgroups of  $S_{12}$  with non-abelian centralizer.

| id | $G$                            | regular? | $ G $ | $ G_1 $ | $C_{S_{12}}(G)$   | mechanism |
|----|--------------------------------|----------|-------|---------|-------------------|-----------|
| 3  | $D_6$                          | yes      | 12    | 1       | $D_6$             | (R)       |
| 4  | $A_4$                          | yes      | 12    | 1       | $A_4$             | (R)       |
| 5  | $C_3 \rtimes C_4$              | yes      | 12    | 1       | $C_3 \rtimes C_4$ | (R)       |
| 16 | $S_3 \times S_3$               | no       | 36    | 3       | $S_3$             | (SN)      |
| 17 | $(C_3 \times C_3) \rtimes C_4$ | no       | 36    | 3       | $S_3$             | (SN)      |
| 35 | $(S_3 \times S_3) \rtimes C_2$ | no       | 72    | 6       | $S_3$             | (SN)      |

## 7. DEGREE 16

The degree  $d = 16$  is the first case in our computations where the number of nonabelian-centralizer exceptions becomes substantially larger. A complete SageMath enumeration of the  $N_{16} = 1954$  transitive subgroups of  $S_{16}$  gives

$$A_{16} = 1929, \quad N_{16} - A_{16} = 25, \quad \frac{A_{16}}{N_{16}} = \frac{1929}{1954}.$$

**Theorem 7.1.** *Among the transitive subgroups of  $S_{16}$ , exactly 25 have non-abelian centralizer. Of these, 9 arise from the regular mechanism, and 16 arise from the stabilizer-normalizer mechanism.*

*Proof.* We use the optimized enumeration described in Section 4. For each transitive subgroup  $G \leq S_{16}$ , the program computes the point stabilizer  $G_1$ , the normalizer  $N_G(G_1)$ , and the quotient

$$N_G(G_1)/G_1.$$

By Proposition 3.1,

$$C_{S_{16}}(G) \cong N_G(G_1)/G_1.$$

Thus  $G$  has nonabelian centralizer exactly when this quotient is nonabelian.

First consider the regular cases. There are exactly 14 groups of order 16 up to isomorphism. Among them, 5 are abelian:

$$C_{16}, \quad C_8 \times C_2, \quad C_4 \times C_4, \quad C_4 \times C_2^2, \quad C_2^4.$$

Thus the remaining 9 groups of order 16 are nonabelian. Each abstract group of order 16 has a regular permutation representation on 16 points, and nonisomorphic abstract groups give distinct  $S_{16}$ -conjugacy classes of regular subgroups. If  $G \leq S_{16}$  is regular, then Proposition 3.2 gives

$$C_{S_{16}}(G) \cong G.$$

Therefore the 9 nonabelian groups of order 16 give 9 regular-mechanism exceptions.

It remains to account for the nonregular cases. The same enumeration checks all nonregular transitive subgroups of  $S_{16}$ . It finds exactly 16 nonregular groups for which

$$N_G(G_1)/G_1$$

is nonabelian. These are the stabilizer-normalizer exceptions. Adding them to the 9 regular exceptions gives

$$N_{16} - A_{16} = 9 + 16 = 25.$$

Since  $N_{16} = 1954$ , this gives

$$A_{16} = 1954 - 25 = 1929.$$

The code used for this enumeration is listed among the computational artifacts in Appendix A.  $\square$

TABLE 4. The nine regular nonabelian transitive subgroups of  $S_{16}$ . SmallGroup IDs refer to GAP's SmallGroups library.

| SmallGroup id | $G$                               | $ G $ | $ C_{S_{16}}(G) $ | centralizer type |
|---------------|-----------------------------------|-------|-------------------|------------------|
| (16, 3)       | $(C_4 \times C_2) \rtimes C_2$    | 16    | 16                | nonabelian       |
| (16, 4)       | $C_4 \rtimes C_4$                 | 16    | 16                | nonabelian       |
| (16, 6)       | $M_{16}$ (modular maximal-cyclic) | 16    | 16                | nonabelian       |
| (16, 7)       | $D_8$ (dihedral)                  | 16    | 16                | nonabelian       |
| (16, 8)       | $SD_{16}$ (semidihedral)          | 16    | 16                | nonabelian       |
| (16, 9)       | $Q_{16}$ (generalised quaternion) | 16    | 16                | nonabelian       |
| (16, 11)      | $D_4 \times C_2$                  | 16    | 16                | nonabelian       |
| (16, 12)      | $Q_8 \times C_2$                  | 16    | 16                | nonabelian       |
| (16, 13)      | $C_4 \circ D_4$ (Pauli)           | 16    | 16                | nonabelian       |

Table 4 lists the nine regular-mechanism exceptions. The remaining sixteen exceptions are nonregular stabilizer-normalizer exceptions.

**7.1. SymPy verification of the regular cases.** We constructed each of the nine regular groups in Table 4 directly as a regular permutation group on 16 points, using its right regular representation. For every  $G$  in the table, `PermutationGroup.centralizer` applied with  $S_{16}$  as the ambient group returned a group of order 16, and `is_abelian` returned `False`. This provides an independent check of the nine regular-mechanism exceptions.

We also computed  $|N_G(G_1)|$  by an explicit conjugation test and verified

$$|C_{S_{16}}(G)| = |N_G(G_1)/G_1|$$

in every case, in agreement with Proposition 3.1.

**7.2. Nonregular exceptions at  $d = 16$ .** The complete SageMath enumeration identifies exactly 16 nonregular transitive subgroups  $G \leq S_{16}$  for which

$$N_G(G_1)/G_1$$

is nonabelian. These are precisely the stabilizer-normalizer exceptions at degree 16. Adding them to the 9 regular exceptions of Table 4 gives the total

$$N_{16} - A_{16} = 25.$$

Each of the 16 nonregular exceptions has  $|G_1| > 1$ , so none comes from the regular mechanism. Thus the degree 16 computation shows that the stabilizer-normalizer mechanism can produce more exceptions than the regular mechanism: at  $d = 16$  the split is

$$16 \text{ SN exceptions} \quad \text{versus} \quad 9 \text{ regular exceptions.}$$

A clean group-theoretic classification of these 16 nonregular exceptions is left for future work.

**7.3. Discussion.** The degree  $16 = 2^4$  suggests that powers of two deserve separate attention. The number of abstract groups of order  $2^n$  grows rapidly with  $n$ , and every nonabelian group of order  $2^n$  contributes a regular nonabelian-centralizer exception in degree  $2^n$ . Thus powers of two naturally produce many regular-mechanism exceptions.

The degree 12 computation suggests a different source of exceptions. There the nonregular exceptions have quotient

$$N_G(G_1)/G_1 \cong S_3.$$

Since  $|S_3| = 6$ , Corollary 3.1 implies that such  $S_3$ -centralizer exceptions can occur only in degrees divisible by 6. This suggests that multiples of 6 may be a natural place to search for further stabilizer-normalizer exceptions.

## 8. FUTURE DIRECTIONS

The computations above raise several natural questions about the structure and frequency of nonabelian centralizers. The prime-degree case has already been settled in Theorem 2.1: if  $p$  is prime and  $G \leq S_p$  is transitive, then  $C_{S_p}(G)$  is either trivial or cyclic of order  $p$ . Thus prime degrees cannot contribute nonabelian-centralizer exceptions.

We record several remaining directions suggested by the data.

*Question 1.* Can the nonregular exceptions, such as the groups with ids 16, 17, 35 at  $d = 12$  and the nonregular exceptions at  $d = 16$ , be classified into an infinite family? The repeated appearance of quotients such as  $S_3$  suggests that stabilizer-normalizer exceptions may admit a uniform description.

*Question 2.* Does the proportion  $A_d/N_d$  tend to 1 as  $d \rightarrow \infty$ ? Equivalently, are transitive groups with nonabelian centralizer asymptotically rare?

*Question 3.* Powers of two appear to produce many regular-mechanism exceptions, because every nonabelian group of order  $2^n$  gives a regular subgroup of  $S_{2^n}$  with nonabelian centralizer. Can the contribution of regular exceptions at degrees  $d = 2^n$  be described explicitly?

*Question 4.* The degree 12 computation suggests that degrees divisible by 6 may be natural sources of stabilizer-normalizer exceptions, since the nonregular exceptions there have quotient  $N_G(G_1)/G_1 \cong S_3$ . Are there systematic families of  $S_3$ -centralizer exceptions in degrees divisible by 6?

## APPENDIX A. COMPUTATIONAL ARTEFACTS

All code accompanying this draft is in the same directory as the source:

- **sage\_verify.sage** — the optimized SageMath driver used to enumerate transitive groups from GAP’s **TransitiveGroups** library, compute  $G_1$ ,  $N_G(G_1)$ , and  $N_G(G_1)/G_1$ , and record the nonabelian centralizer exceptions. This script produces **summary.csv** and the per-degree exception tables, including the degree-12 enumeration used in Theorem 6.1.
- **sympy\_verify.py** — the independent SymPy re-check for selected centralizer computations, including the regular cases at  $d = 12$  and  $d = 16$ .

Table 5 records the actual wall-times measured on a single laptop core (the same machine that produced Table 2). The total wall-time for the entire sweep  $4 \leq d \leq 30$  is under six minutes, dominated by the single row  $d = 24$ .

TABLE 5. Wall-time of the optimised pipeline (Listing 2) on a single 2024-era laptop core, per degree.

| $d$ range            | wall-time (improved pipeline, single laptop core)     |
|----------------------|-------------------------------------------------------|
| $4 \leq d \leq 15$   | < 4 s total                                           |
| $d = 16$             | 11.1 s                                                |
| $d = 17, 19, 23, 29$ | < 10 ms each (prime-degree theorem)                   |
| $d = 18$             | 6.3 s                                                 |
| $d = 20$             | 8.8 s                                                 |
| $d = 21, 22, 25, 26$ | 0.4–1.2 s each                                        |
| $d = 24$             | 225.2 s ( $\approx 3.8$ min)                          |
| $d = 27$             | 17.2 s                                                |
| $d = 28$             | 17.4 s                                                |
| $d = 30$             | 56.3 s                                                |
| $d = 32$             | out of reach (memory: $\sim 2.8 \times 10^6$ classes) |

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